

Response to Reviewers [HIGHLIGHTED VERSION]

Knowledge-Based Systems Elsevier

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*Effective Training Pressure Gates Recursive Knowledge Degradation
in LLMs: A Multi-Axis Dose-Response Study*

Revision round: 1 Date: October 4, 2026

We thank the editor and both reviewers for their careful reading of the manuscript. Each point has been addressed systematically below. Changes to the manuscript text are shown in *blue italic* when quoted in this letter and highlighted in blue in the marked-up manuscript version.

Legend (highlighted version): inserted / changed text ~~deleted text~~ *text quoted from manuscript*

Editor

Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.

We have revised the manuscript throughout for clarity and have used Elsevier’s Author Services English Language Editing service. The point-by-point response follows.

1 Reviewer 1

R1.Q1

How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?

We agree that the present experiments do not establish a transferable scalar ETP predictor or a prospective threshold for an unseen backbone or dataset. We have revised the manuscript to distinguish controllable design variables—nominal adapter rank, learning rate, planned training steps, and synthetic-data dose—from post-training diagnostics, including effective rank and weight drift. Under the tested protocol, we identify empirical intervals between adjacent configurations exhibiting different retention trajectories; these intervals should not be interpreted as universal numerical thresholds.

A prospective test would require: (i) specifying a calibration rule and outcome criterion before examining a held-out backbone or dataset; (ii) selecting configurations on both sides of the predicted regime boundary; and (iii) evaluating them without refitting that rule to the held-out outcomes. We have added this as a concrete validation protocol in Section ?? and state explicitly that such held-out prospective validation has not been performed in the present study. Accordingly, we describe effective training pressure as an operational experimental framework, not a validated universal predictor.

The pre-registered G5 experiment (three cells, one seed) provides exploratory evidence that the operational regime index correctly predicts qualitative behavior (homeostatic vs. bounded) for pre-specified configurations on the tested backbone. Prediction of the exact degradative threshold remains uncalibrated.

Location in revised manuscript: *Section 3.5 (ETP as framework, revised); Section ?? (prospective calibration protocol); Section 4.4 (G5 prospective results).*

R1.Q2

How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?

We thank the reviewer for this precise question. The revised manuscript addresses each potential confound through a dedicated control, all of which are now documented in Section 3.9 (“Sources of variation and experimental controls”, ll. 1000–1037) with Table 2 (tab:controls).

Quantization (NF4 vs bf16). All QLoRA experiments in the main analysis use 4-bit NormalFloat (NF4) quantization of the base weights, with adapters trained in bf16. A dedicated ablation (G3, three seeds each at $r = 16$ and $r = 256$) runs the same protocol with standard bf16 precision and no base-weight quantization. G3 results show that the regime classification is preserved under both precision settings: at $r = 16$ bf16 achieves Gen10=97.4%, identical to the NF4 baseline; at $r = 256$ bf16 achieves Gen10=79.1%, within 0.0 pp of the NF4 condition. The delta between NF4 and bf16 is ≤ 0.1 pp at both ranks. Quantization is therefore not driving the observed regime separation.

Location in revised manuscript: *Section 3.9 (tab:controls); Section 4.6 (G3 results).*

Optimizer state. A fresh optimizer is instantiated at every generation; no momentum or adaptive-rate state is carried forward between generations. This is held constant across all conditions within a backbone and cannot explain differences between conditions that share the same protocol.

Location in revised manuscript: *Section 3.9 (tab:controls), “Optimizer state” row.*

Training data order. Ablation G4 runs Qwen at $r = 256$ with three independent shuffle seeds (42/101/202), keeping all other conditions identical to G1 (seeds 15/137/256). A permutation test comparing Gen10 retention between the two seed groups gives $p = 0.70$ (two-sided, $N = 3 + 3$), indicating that the data-order variance is statistically indistinguishable from the seed-to-seed variance already present within G1. The combined $N = 6$ seed set (G1+G4) yields Gen10 = $79.7\% \pm 1.7$ pp, with variance attributable to seed rather than order.

Location in revised manuscript: *Section 3.9 (tab:controls); Section 4.6 (G4 results).*

ETP-design vs ETP-observed. Following the reviewer’s implicit distinction, we now separate two classes of ETP variables in the revised manuscript. ETP-design variables are known before training: nominal rank (r), learning rate (η), number of training steps, and synthetic exposure fraction. These are the directly manipulated axes in our factorial design. ETP-observed variables are measured after training: effective rank (erank), mean absolute weight drift ($d(t)$), response length, and SDI-3. We report erank and drift as post-training diagnostic proxies that track how much pressure was actually applied, not as pre-specified predictors. The revised text makes this distinction explicit and avoids treating erank as a prospective input.

Location in revised manuscript: *Section 3.5 (ETP definition); Table 1 (tab:notation).*

Seed-level contrast for the primary headline claim. For the central Qwen result, the $r = 16$ condition is represented by the three G3/bf16 runs (seeds 15/137/256, Gen10 = 76/78 per seed = 97.4%) and the $r = 256$ condition by six G1+G4 runs (seeds 15/137/256/42/101/202, Gen10 = 61, 63, 61, 64, 61, 63 out of 78; mean $79.7\% \pm 1.7$ pp). Seeds 15, 137, and 256 appear in both groups, which differ simultaneously in rank, quantization, and data-order protocol. The

17.7 pp arithmetic gap is therefore reported as a descriptive observation, not as a confirmatory permutation test. The cleanest rank-only comparison is the G3 within-protocol design (same seeds, same bf16 protocol, rank the only variable): per-seed Gen10 differences are +15, +13, and +15 items out of 78, a mean of +18.4 pp; sign-flip $p = 0.25$ ($N = 3$, all positive direction but underpowered).

For Gemma 3, the $r = 4$ versus $r = 16$ Gen10 contrast ($N = 5 + 5$, five seeds per rank) gives $\Delta = +37.8$ pp (from raw counts, $K_0 = 46$). If the same five seeds were used at both ranks (paired design), the exact sign-flip test gives $p = 0.0625$ two-sided ($p = 0.031$ one-sided pre-specified); if seeds are independent, a bilateral permutation test over $\binom{10}{5} = 252$ allocations gives $p = 0.0079$. The effect is consistent and large in either case; we report both and clarify the seed assignment in the revised manuscript.

Location in revised manuscript: *Section 4.2 (Qwen, G3 paired contrast, descriptive gap 17.7 pp); Section 4.3.1 (Gemma 3, $\Delta = +37.8$ pp, $p = 0.0625$ sign-flip or $p = 0.0079$ unpaired permutation).*

R1.Q3

Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?

We thank the reviewer for this question, which targets the external validity of our findings. We address the three distinct concerns it raises: the evaluation metric, the dataset scope, and the extension to other domains.

Evaluation metric and verbosity bias. The primary metric uses a bidirectional substring criterion: a generated response is counted as correct if it contains any valid answer alias, or if any alias contains the response. This criterion is intentionally permissive to handle short-answer variation and aliasing in TriviaQA-style factoid questions. However, as the reviewer correctly anticipates, configurations that generate longer responses may receive inflated retention scores because a longer string is more likely to contain a short alias by chance.

To address this concern, we performed a strict exact-match re-evaluation on the saved predictions for all reported conditions. Under the strict scorer, absolute retention values are lower, but the direction of every regime classification is unchanged: configurations classified as Homeostatic, Bounded, or Degradative retain those classifications under strict scoring. This re-evaluation is reported in supplementary Table S2 and is discussed in Section 6 (ll. 2181–2193).

Location in revised manuscript: *Section 6, ll. 2181–2193; supplementary Table S2.*

Dataset scope: single dataset, short factoid answers. TriviaQA was chosen specifically to isolate factual retention from reasoning or stylistic confounds. Short answers minimise format-dependent evaluation noise, and the substring criterion is resilient to verbosity changes. These are methodological advantages for studying the retention question in isolation, not limitations of the phenomenon. The limitation is that the findings may not transfer to long-form generation, multi-hop reasoning, open-ended dialogue, or code synthesis, where output structure and evaluation criteria differ substantially. This scope restriction is stated explicitly in Section 6 (ll. 2094–2100).

Location in revised manuscript: *Section 6, ll. 2094–2100.*

Larger factual evaluation sets. The held-out evaluation set contains 200 questions, of which $K_0 \in \{78\text{--}79\}$ are answered correctly at generation 0 for the Qwen backbone. [This evaluation set is disjoint from the 2,000-question synthetic training stream](#): the 200 evaluation questions

are held out and never used for synthetic-data generation or training (Section 3.1, ll. 521–523). Scaling the evaluation set would require a proportionally larger source corpus and would alter the training-data composition; it is therefore deferred to future work that also scales the training protocol. Extending to a larger evaluation set is a natural direction for future work and is mentioned in Section 6. The current set is consistent with comparable prior studies [? ?] and is sufficient to characterise regime-level differences across conditions.

Location in revised manuscript: *Section 6, ll. 2094–2100; Section 3.1 (evaluation protocol).*

R1.Q4

Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?

We thank the reviewer for this question, which led directly to a substantially revised and pre-registered experiment. We address three issues in turn: the correction of the original result, the pre-registered dose–response, and the mechanism.

Correction of the original result. The original submission reported that a $\sim 5\%$ reduction of synthetic training examples raised retention by +9.0 and +9.4 pp for conditions C3 and C5 respectively. We subsequently identified that C1 (the baseline) and C5 (the intervention) had been produced by different evaluation pipelines, which differed in the K_0 construction procedure and the data-loading code path. When C1 and C5 are re-run under an identical pipeline with paired seeds, the effect of $\sim 5\%$ example removal is only +1.9 pp (not distinguishable from zero at $n = 3$). This discrepancy is acknowledged explicitly in the revised manuscript (ll. 1708–1713) and the original C2–C5 results are preserved in Appendix A for transparency. We thank the reviewer, whose question prompted the investigation that uncovered this error.

Pre-registered dose–response (Bloco 7). To replace the discredited result, we designed and pre-registered a dose–response experiment (protocol committed as [a916ebf](#) before any run), varying synthetic exposure at 0%, 10%, 25%, and 50% removal across five paired seeds on Qwen 2.5 1.5B at $r = 256$. The results, reported in Section 4.5 (ll. 1685–1713), show that:

- At Gen 5, the gains are +3.6 pp (10%, $p = 0.009$), +3.8 pp (25%, $p = 0.003$), and +4.4 pp (50%, $p < 0.001$), $n = 5$ seeds each; the Gen 5 benefit is uniform across doses while the Gen 10 benefit is dose-graded.
- At Gen 10, the benefit is dose-graded: +1.8 pp at 10% ($p = 0.160$, not significant), +4.9 pp at 25% (95 % CI [2.5, 7.2], $p = 0.005$), and +9.2 pp at 50% (95 % CI [6.9, 11.6], $p < 0.001$).
- At 50% removal the Gen 5→10 slope is -0.15 pp/generation versus -1.13 pp/generation at 0% (mean per-seed difference of Gen10–Gen5 counts divided by five generations), strongly attenuating progressive loss and shifting the system toward the Bounded regime.

The pre-registered prediction that 50% removal would match halving the learning rate (from $\eta = 10^{-5}$ to 5×10^{-6}) yielded a similar point estimate: Gen 5 retention at dose = 50% is 89.7% versus 90.2% at $\eta = 5 \times 10^{-6}$ (mean over three seeds; Gen 10 difference +0.4 pp, 90 % CI [−0.8, 1.6]). Formal equivalence was not established.

Cross-backbone replication and mechanism. A parallel pre-registered dose–response experiment on Gemma 3 1B IT at $r = 10$ (G2, three paired seeds, $K_{0,G2} = 44$) shows that a 50% reduction raises Gen 10 retention by +12.9 pp (95 % CI [9.70, 16.10], $p = 0.001$). However, the Gen 5→10 slope in the G2 50% arm (-0.91 pp/generation) is equal to that of the 0% arm (-0.91 pp/generation) within rounding, indicating that the exposure effect on Gemma 3 manifested as a Gen 10 retention gain rather than as a reduction in the rate of progressive loss. The mechanism was not established. A steps-matched control was inconclusive at $n = 3$. The overall

finding is: on Qwen B7, the 50% dose strongly attenuated progressive loss; on Gemma 3 G2, it improved Gen 10 retention without reducing the observed rate of loss.

Location in revised manuscript: *Section 4.5 (B7 Qwen and G2 Gemma 3 dose-response; pre-registration; LR equivalence); Section 6 (mechanism limitation, Gemma 3 r=16 pilot); repository (original C1–C5 pilot archived).*

R1.Q5

What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?

We agree that independent reproduction requires more than aggregate retention tables. We have expanded the reproducibility section (Section ??) to specify:

1. the exact model checkpoint identifiers and revisions for all three backbones;
2. train and evaluation question IDs, and the baseline-correct K_0 sets ($|K_0| = 78$ for Qwen; two Gemma 3 sets: $|K_{0,\text{rank}}| = 46$ for rank-sweep experiments and $|K_{0,\text{G2}}| = 44$ for dose-response G2; see Section 3.6);
3. prompt and chat templates used for both synthetic data generation and deterministic evaluation;
4. distinct decoding settings (sampling parameters for generation, `do_sample=False` and `temperature=0` for evaluation);
5. correctness normalization and alias-expansion code;
6. training configurations, software versions, random seeds, data-order and downsampling masks;
7. per-generation retention counts and boolean correctness vectors (length $|K_0|$ per seed and generation) are included in the consolidated CSV; raw token-level predictions are not in the current release but available from the authors upon request; and
8. scripts used to regenerate every table and figure.

We also specify which experiments reset adapters and optimizer states between generations. The replication package contains: (1) all analysis and figure-generation scripts; (2) the consolidated results CSV (G1–G5, 27 executed training runs, per-generation per-seed retention); (3) all 11 publication figures at 300dpi; and (4) the pre-registration commit for B7 ([a916ebf](#)). The package does not contain raw token-level predictions, adapter checkpoints (available on request; ≈ 70 MB each in bf16), or base-model weights (obtained from their original distributors). It is archived at <https://doi.org/10.5281/zenodo.23146342>. Release v1.0.3 contains the initial replication scripts and data; the revised analysis scripts and updated `numbers.tex` from the current submission are not yet included and will be deposited in a subsequent release. Running `python scripts/make_all.py` reproduces all 112 canonical numbers (15/15 checks pass). An artifact manifest with SHA-256 hashes is in `MANIFEST.md`. Base-model weights are obtained from their original distributors and are not redistributed. The revised appendix provides an artifact manifest mapping every reported figure and table to its generating data and script.

Location in revised manuscript: *Section ?? (repository and data access, expanded).*

2 Reviewer 2

R2.1 — Abbreviations and notation

ETP appears in Figure 1 (“ETP threshold identification”) but is never explicitly defined as “effective training pressure.”

We agree. The revised manuscript does not use “ETP” as a standalone acronym. The term is introduced by name at l. 72 (“*which we term effective training pressure*”) and used in full throughout. Figure 1 (caption ll. 692–698) no longer contains the abbreviation in isolation.

Location in revised manuscript: *Abstract l. 72; Figure 1 caption ll. 692–698.*

MTLD is introduced as “a length-robust diversity estimate” but the acronym is not expanded.

Corrected. The first occurrence now reads “*Measure of Textual Lexical Diversity (MTLD)*” followed by the functional description.

Location in revised manuscript: *l. 875.*

KL, JS, and coverage appear in Figure 1 without definition.

These three quantities were used as conceptual labels for distribution-comparison diagnostics that are not computed or reported anywhere in the manuscript. They have been removed from Figure 1 entirely. The revised caption describes only quantities that are measured and reported: factual retention, effective rank, and distributional shift (response length, content efficiency, SDI-3).

Location in revised manuscript: *Figure 1 caption, ll. 692–698.*

bf16, SVD, TCE, QLoRA, and LoRA are used without consistently expanded forms.

All four acronyms now have explicit parenthetical expansions at first use: “*full fine-tuning (FFT)*” at l. 75; “*singular value decomposition (SVD)*” at l. 332; “*Quantized Low-Rank Adaptation (QLoRA)*” at l. 333. The term *bf16* (BFloat16) is glossed on first use. TCE does not appear in the revised manuscript. LoRA was already expanded in the Related Work section.

Location in revised manuscript: *ll. 75, 332, 333.*

“Homeo.” is used in Table 2 without definition.

The abbreviation has been expanded to “*Homeostatic*” in all four occurrences within the rank-sweep results table.

Location in revised manuscript: *Table 2 (tab:rank_sweep), regime column.*

K_0 is used both as a set of questions and as a retention metric. This dual use is not always clear.

We have added a dedicated notation table (Table 1, ll. 438–445) that distinguishes the two uses explicitly: K_0 denotes the integer cardinality of the held-out question set answered correctly at generation 0; $R(t) = K_t/K_0$ denotes the retention ratio at generation t . The symbols $R(t)$, $d(t)$, and the efficiency numerator have been harmonised throughout Equations 1–6 and the surrounding prose.

Location in revised manuscript: Table 1 (tab:notation), ll. 438–445; Equations 1, 3, 5.

T , $D_{1,T}$, D_{10} , ΔI , $\text{Instability}(1)$, MeanLen_T , and related symbols in Eq. 6 are not fully defined.

All symbols in Eq. 6 (SDI-3) and its supporting quantities are now defined in Table 1 (tab:notation) and in the inline text preceding the equations: T = number of generations evaluated; $D_{1,T}$ and D_{10} = content efficiency at generation 1 and generation 10; ΔI = absolute change in content efficiency over the evaluation horizon; $\text{Instability}(1)$ = SDI-3 sub-score at the first generation; MeanLen_T = mean response length averaged over all T generations.

Location in revised manuscript: Table 1 (tab:notation); prose preceding Eq. 6.

The effective-rank formula uses σ_i as both singular values and normalised singular values. The normalisation condition is mathematically incorrect as written.

Corrected. Eq. 2 now defines $p_i = \sigma_i / \sum_j \sigma_j$ explicitly, reserving σ_i for the raw singular value throughout. The entropy is computed over the normalised distribution $\{p_i\}$, consistent with Table 1.

Location in revised manuscript: Eq. 2 (erank definition); Table 1 (tab:notation).

$\|BA\|_F$, $\theta_i^{(t)}$, $\theta_i^{(0)}$, and related notation in the drift equation are not consistently defined.

Both quantities are now defined in Table 1 and at first use: $\|BA\|_F$ denotes the Frobenius norm of the LoRA reconstruction matrix, averaged across adapted modules; $\theta_i^{(t)}$ and $\theta_i^{(0)}$ denote the i -th parameter at generation t and at generation 0. The mean absolute drift $d(t)$ in Eq. 3 is the mean over all i of $|\theta_i^{(t)} - \theta_i^{(0)}|$.

Location in revised manuscript: Table 1 (tab:notation); Eqs. 3–4 and surrounding prose.

“ $r \in \{4, 10, 12, 14, 16\}$ ” and “ $r \approx \{10, 12, 14\}$ ” are garbled in the PDF; they should use clear set notation.

The garbled rendering was a font-encoding artefact in the PDF generated from the original submission. The revised manuscript uses the `lmodern` package with T1 encoding throughout, which resolves all set-notation rendering. The expressions now compile correctly as $r \in \{4, 10, 12, 14, 16\}$ and $r \in \{10, 12, 14\}$.

Location in revised manuscript: Preamble (`lmodern + T1`); all set-notation instances throughout.

R2.2 — Equations and tables

Equations 1–6 are severely malformed in the provided PDF. Eq. 1 uses “Retention.t /” instead of a clear function notation and has a garbled summation over K_0 .

The malformed rendering was an artefact of the font encoding used in the original submission: Unicode characters in mathematical mode were not mapped correctly to the T1 font metric, producing garbled output in the compiled PDF while the source was syntactically correct. The revised submission uses the `lmodern` package with T1 encoding throughout (preamble), which eliminates the rendering artefact.

Eq. 1 now reads correctly as

$$R(t) = \frac{|\{q \in K_0 : q \text{ correct at gen. } t\}|}{|K_0|},$$

using $R(t)$ as the retention symbol, consistent with Table 1 (tab:notation) and the surrounding prose. The symbol $\text{Retention}(t)$ that appeared in the original version has been replaced by $R(t)$ throughout Equations 1, 3, and 5 and the accompanying text.

Location in revised manuscript: *Preamble (lmodern + T1); Eq. 1, l. 595.*

Eq. 2 for effective rank is missing the negative sign and summation. It should be $\exp(-\sum_i p_i \log p_i)$, with $p_i = \sigma_i / \sum_j \sigma_j$.

Corrected. Eq. 2 now reads

$$\text{erank}(BA) = \exp\left(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\right),$$

where $\hat{\sigma}_i = \sigma_i / \sum_j \sigma_j$ are the normalised singular values of BA . The normalised symbol $\hat{\sigma}_i$ is distinct from the raw singular value σ_i , consistent with Table 1. The formula is the standard effective-rank definition of ?].

Location in revised manuscript: *Eq. 2, ll. 798–803.*

Eq. 6 for SDI-3 is unclear: signs, log ratios, and the ΔI term are not consistently defined.

Eq. 6 now reads

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I,$$

with each term defined immediately before the equation and in Table 1: the first term is positive when mean response length grows over the evaluation horizon; the second is positive when content efficiency (D1) declines (the numerator and denominator are swapped relative to the length term, encoding the opposite direction of “worse”); and ΔI is the absolute drop in content efficiency from generation 1 to generation T . A positive SDI-3 score therefore indicates cumulative distributional deterioration.

Location in revised manuscript: *Eq. 6, ll. 881–886; Table 1 (tab:notation); prose ll. 870–890.*

What is “ 5×10^6 ” in Table 7? Please check also elsewhere.

The garbled rendering was again a font-encoding artefact. The value in Table 7 (tab:ranklr) is the learning rate 5×10^{-6} , not 5×10^6 . The exponent -6 was dropped in the original PDF due to the same encoding issue. The revised PDF renders the value correctly as 5×10^{-6} throughout (Table 7 header row, l. 1473, and all inline references). We have verified that no other table or equation contains a similar rendering failure in the revised submission.

Location in revised manuscript: *Table 7 (tab:ranklr), l. 1473; preamble (lmodern + T1).*

R2.3 — Technical and data inconsistencies

We thank the reviewer for this careful cross-checking of results against tables and prose. Each inconsistency has been corrected as described below.

Qwen threshold inconsistency. Abstract and Section 4.2.3 state the Qwen boundary is between effective ranks 50–88. Section 5.1 states Qwen remains outside the degradative regime “up to effective rank 150.” This contradicts Table 2, where $r = 256$ has effective rank 87.57 and is classified as degradative.

The phrase “up to effective rank 150” has been removed. All threshold language now uses the form “*an abrupt, threshold-like transition within the tested grid*” (l. 72, l. 95), which accurately reflects that the boundary is established empirically between adjacent tested configurations rather than extrapolated to untested values. The interval 50–88 (effective ranks of $r = 128$ and $r = 256$ respectively) is retained as the localization of the boundary. No claim is made beyond the tested range.

Location in revised manuscript: Abstract l. 72; Contribution #1 l. 95; Section 5.1 (phrase removed).

$r = 128$ classification. $r = 128$ is classified as “Bounded” in Table 2, but Section 4.4.3 calls $r = 128$ and $r = 256$ “above-threshold configurations.”

The phrase “above-threshold configurations” in Section 4.4.3 now refers only to $r = 256$, which Table 2 classifies as Degradative. The term “*above-threshold*” is used exclusively for configurations in the Degradative regime. $r = 128$ (Bounded) is described as a “*transition-zone configuration*” where distributional signatures begin to emerge without producing the full degradative trajectory.

Location in revised manuscript: Section 4.4.3, ll. 1061, 1570.

$r = 256$ boundary claim. $r = 256$ is repeatedly called a “boundary configuration,” but Table 2 classifies it as degradative. The boundary lies between $r = 128$ and $r = 256$.

Agreed. The five occurrences of “boundary configuration” referring to $r = 256$ have been replaced throughout with “*lowest-pressure degradative configuration tested*”, which correctly identifies $r = 256$ as the degradative anchor while making explicit that the transition itself is localised between $r = 128$ and $r = 256$.

Location in revised manuscript: ll. 1737, 1915, 2063, 2069, 2159.

Gemma 3 degradative range. Section 5.2 states “ $r \geq 16$ on Gemma 3,” but Table 4 shows $r \in \{10, 12, 14\}$ already degradative at Gen 5. The boundary is below $r = 10$.

Corrected. Table 4 (tab:gemma3) reports retention of 78.3%, 73.9%, and 69.6% for $r = 10, 12$, and 14 respectively at Gen 5, all classified as Degradative. The regime boundary is placed at effective ranks between 3 and 6 (between $r = 4$, which is Homeostatic, and $r = 10$, which is Degradative). The text now reads “*the Gemma 3 regime boundary lies between effective ranks 3 and 6, below $r = 10$* ”, consistent with Table 4. The phrase “ $r \geq 16$ ” has been removed.

Location in revised manuscript: Section 5.2; Table 4 (tab:gemma3), ll. 1265, 1277–1280.

Abstract overclaim. “Three to five independent seeds per condition” is not accurate: many conditions are single-seed; only headline comparisons use multiple seeds.

The abstract now reads “*three to five independent seeds for headline conditions (single seed for intermediate ablations)*” (l. 69), which accurately reflects the experimental design. Headline rank levels ($r \in \{16, 256\}$ on Qwen; $r \in \{4, 16\}$ on Gemma 3) use three to five seeds; intermediate and ablation conditions use a single seed. This distinction is also stated in the Methods and

reported per table via the added N column.

Location in revised manuscript: *Abstract l. 69.*

FFT on Gemma 3: non-monotonic results. Retention values of 78.3 %, 54.3 %, and 63.0 % for learning rates 10^{-6} , 5×10^{-6} , and 10^{-5} are non-monotonic. The conclusion that this “confirms” the magnitude axis is too strong for a single-seed, non-monotonic result.

Agreed. The text (l. 1147–1152) now notes explicitly that “*no rank reversal or non-monotonicity is observed*” in the primary Qwen rank sweep; for the FFT on Gemma 3, the word “confirms” has been replaced with “*is consistent with*” and the non-monotonic pattern is acknowledged: the result is treated as supporting evidence at $n = 1$ seed rather than confirmatory.

Location in revised manuscript: *ll. 1147–1149 (Qwen rank sweep prose); FFT Gemma 3 paragraph.*

C5 removal percentage. Section 3.4 mentions “15% removal,” while Section 4.5 reports approximately 5 % token reduction. This discrepancy must be clarified.

The two figures refer to different quantities. The original C5 condition removed 15 % of training examples (questions), which reduced the token count by approximately 5 % because responses vary in length. The revised manuscript clarifies this distinction at both locations. However, the more substantive resolution is that the original C3/C5 results were produced by a different evaluation pipeline than C1: when C1 and C5 are re-run under an identical pipeline with paired seeds, the effect of $\sim 5\%$ example removal is +1.9 pp (not distinguishable from zero, $n = 3$). The dose–response reported in Section 4.5 (Bloco 7) replaces those comparisons; the original C2–C5 results are preserved in Appendix A for transparency.

Location in revised manuscript: *Section 3.7 (definition of C5 doses); Section 4.5 l. 1708–1713.*

R2.4 — Methodology

We thank the reviewer for this detailed methodological critique. We address each concern in turn.

No formal statistical tests, confidence intervals, or p-values are reported. The paper relies on ranges and non-overlap, which is weak for inference.

We agree. The revised manuscript adds seed-level exact permutation tests for the comparisons where independent training runs are available, treating seeds/runs as the experimental unit.

The primary rank contrast for Qwen uses three G3/bf16 runs at $r = 16$ (seeds 15/137/256; Gen10 = 76/78 per seed = 97.4%) and six G1+G4 runs at $r = 256$ (seeds 15/137/256/42/101/202; Gen10 = 61, 63, 61, 64, 61, 63 out of 78; mean $79.7\% \pm 1.7$ pp; range 78.2–82.1%). We report this 17.7 pp arithmetic gap as a descriptive observation. Because seeds 15, 137, and 256 appear in both groups, and the groups differ simultaneously in rank, quantization level, and data-order protocol, the nine observations are not exchangeable under the rank-only null hypothesis; we therefore do not report a formal permutation test for this comparison.

The cleanest available within-protocol rank comparison is the G3 paired design: same three seeds (15/137/256), same bf16 protocol, rank the only variable. Per-seed Gen10 differences are +15, +13, and +15 items out of 78, a mean of +18.4 pp. An exact bilateral sign-flip test gives $p = 0.25$ owing to $N = 3$, despite consistent effect direction across all three pairs. This is described as a large but underpowered result.

For Gemma 3, the $r = 4$ versus $r = 16$ Gen10 contrast ($N = 5 + 5$, fully independent seeds, $K_0 = 46$) gives $\Delta = +37.8$ pp (computed from raw counts); an exact bilateral permutation test over $\binom{10}{5} = 252$ allocations gives $p = 0.0079$.

For the dose-response experiments, paired t -tests are reported per dose level against the unmodified baseline, with p -values and 95 % confidence intervals for the +9.2 pp (50%) and +4.9 pp (25%) Gen10 estimates (Section 4.5).

We do not treat the $K_0 = 78$ or $K_0 = 46$ evaluation items as independent binomial trials. Item-level analyses are presented as descriptive summaries of which individual facts were lost or recovered.

Location in revised manuscript: *Section 4.2 (Qwen, G3 paired contrast $N = 3$, descriptive gap 17.7 pp); Section 4.3.1 (Gemma 3, $\binom{10}{5} = 252$, $p = 0.0079$); Section 4.5 (dose-response, paired t -test).*

The effective training pressure framework is not quantified. It is a useful organizing metaphor, but not a predictive model.

We agree with this characterisation and have revised the manuscript accordingly. ETP is now presented as an operational experimental framework rather than a validated scalar predictor. We distinguish two classes of variables: ETP-design (rank, learning rate, exposure fraction, number of steps – variables known before training and directly manipulated) and ETP-observed (erank, drift, response length – post-training diagnostics that track the pressure actually applied). The product $\text{erank} \times \text{LR}$ is reported as an exploratory index and labelled as such; it is not used as a prospectively validated predictor.

Location in revised manuscript: *Section 3.5; Table 1 (tab:notation).*

Cross-architecture normalisation failed, but no alternative calibration method is proposed.

We agree that the failed normalisations in Table 6 (tab:normalization) do not provide a universal scalar. In the revised manuscript we describe this as a limitation: architecture-specific threshold calibration requires either a held-out backbone or a pre-specified normalization rule with out-of-sample validation. We do not propose a universal ETP constant. The table is retained as evidence of incommensurability rather than as a calibration tool.

Location in revised manuscript: *Section 6 (Limitations); Table 6 (tab:normalization).*

“Effective training pressure” is not a new quantitative measure; it is a conceptual label for three already-known axes. The paper does not derive a scalar predictor or formal theory.

We accept this characterisation. The revised Introduction and Contribution #1 now describe ETP as a factorial organisational framework that groups three controllable axes under a common label, enabling joint dose-response analysis. We do not claim to derive a new quantitative measure. The contribution is the factorial design, the dose-response evidence across three backbones, the pre-registered exposure result, and the identification of architecture-specific regime boundaries, not the label itself.

Location in revised manuscript: *Section 1 (Introduction, revised Contribution #1).*

The Gemma 4 E2B backbone lacks a citation, may be confused with Gemma 3n E2B, and its architecture is described without a source.

Corrected. The backbone is now cited as ?] (Gemma Team, 2026, arXiv:2607.02770) at every

introduction point. The architecture details (hidden size 1536, 35 layers, 8 attention heads, 1 KV head, `num_kv_shared_layers`= 20, 50 adapted LoRA modules) are sourced from the released model configuration in the same technical report. The model is unambiguously distinct from Gemma 3n E2B as it uses a different parameter count, layer depth, and attention configuration.

Location in revised manuscript: *Section 3.3 (Models and hardware), ll. 647–662.*

The “sharp transition” claim is weakened by sparse rank grids and wide threshold intervals.

Agreed. The revised manuscript replaces “sharp transition” with “*abrupt, threshold-like transition within the tested grid*”, acknowledging that the transition is characterised empirically by the interval between adjacent tested rank values rather than by a precisely identified point. We report threshold intervals rather than point estimates and describe the intermediate rank sweep ($r \in \{10, 12, 14\}$ on Gemma 3) as a localisation of the transition region, not a sharp boundary identification.

Location in revised manuscript: *Abstract l. 72; Contribution #1 l. 95; Section 4.2.3.*

R2.5 — Format and citations

We thank the reviewer for auditing the reference list. Each issue has been corrected as described below.

References [12] and [33] share the same arXiv number: arXiv:2408.14572.

In the revised reference list the arXiv number 2408.14572 appears exactly once, assigned to ?] (*CURLoRA*, Fawi, 2024). The duplicate entry that appeared in the submitted version has been removed. The reference formerly numbered [33] in the submitted PDF was an artefact of a bibliography management error; it is no longer present.

Location in revised manuscript: *References section (cas-refs.bib), entry fawi2024curlora.*

Reference [32] lists arXiv:2512.15634 with year 2024, which is impossible if the arXiv ID is from December 2025.

Corrected. The entry `ding2024ranktrade` (Rathore et al.) now carries `year = {2025}`, consistent with the December 2025 arXiv submission date. The paper appeared in the proceedings of AACL-IJCNLP 2025. The `year = {2024}` in the submitted version was a transcription error introduced when the preliminary arXiv preprint was first catalogued.

Location in revised manuscript: *References section, entry ding2024ranktrade.*

No citation is provided for Gemma 4 E2B IT, despite it being a central backbone.

A full citation has been added. The backbone is now cited as ?] (Gemma Team, 2026, arXiv:2607.02770) at every point where it is introduced or described, including the experimental setup (Section 3.3) and the cross-backbone results (Section 4.3). The model architecture details (hidden dimension, layers, LoRA target modules) are sourced from the same technical report.

Location in revised manuscript: *Section 3.3; Section 4.3; References entry gemma4_2026.*

Reference [7] “Keisha et al.” is used heavily but appears to be a preprint with no venue. Its reliability should be clarified.

The entry `keisha2025` (Keisha et al., 2025) is a non-peer-reviewed preprint (arXiv:2509.04796). This is now declared explicitly in the bibliography note and in the text at first citation: “[? \] \(non-peer-reviewed preprint\) report a complementary observation...](#)”. The paper is used for contextual framing only; no quantitative claim in our manuscript depends exclusively on this source. We have added `? ]` (published in *Nature*) as the primary citation for the model-collapse phenomenon wherever both sources were previously cited.

Location in revised manuscript: *First citation of keisha2025; References entry with note = {arXiv:2509.04796. Non-peer-reviewed preprint}.*

Reference formatting is inconsistent: some entries include conference proceedings, others only arXiv, some lack page numbers or venues.

The reference list has been audited and harmonised. Published works include their full venue (journal or conference proceedings, volume, pages, DOI where available). Works that exist only as arXiv preprints are formatted as `arXiv preprint arXiv:XXXX.XXXXX` with the submission year. Entries that had previously been submitted with incomplete venue data – including `ding2024ranktrade`, `fawi2024curlora`, and `zibakhsh2024` – have been completed. The full revised bibliography is provided as `cas-refs.bib` in the submission package.

Location in revised manuscript: *References section (cas-refs.bib), all entries.*